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September 20, 2017

L-MT-17-068
10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Monticello Nuclear Generating Plant
Docket No. 50-263
Renewed Facility Operating License No. DPR-22

LER 2017-005-00, "Diesel Generator Emergency Service Water System Automatic Transfer to Alternate Shutdown Panel"

Enclosed is the Monticello Nuclear Generating Plant (MNGP) Licensee Event Report (LER) LER 2017-005-00, "Diesel Generator Emergency Service Water System Automatic Transfer to Alternate Shutdown Panel". This condition is reportable to the NRC in accordance with 10 CFR 50.73(a)(2)(iv)(A), as a condition that resulted in the automatic actuation of the Emergency Service Water System.

Summary of Commitments

This letter makes no new commitments and no revisions to existing commitments.

Christopher R. Church
Site Vice President, Monticello Nuclear Generating Plant
Northern States Power Company – Minnesota

Enclosure

cc: Administrator, Region III, USNRC
Project Manager, Monticello Nuclear Generating Plant, USNRC
Resident Inspector, Monticello Nuclear Generating Plant, USNRC

NRC FORM 366 (04-2017) U.S. NUCLEAR REGULATORY COMMISSION	APPROVED BY OMB: NO. 3150-0104 EXPIRES: 03/31/2020
LICENSEE EVENT REPORT (LER) (See Page 2 for required number of digits/characters for each block) (See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)	Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the Information Services Branch (T-2 F43), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects. Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

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4. TITLE
Diesel Generator Emergency Service Water System Automatic Transfer to Alternate Shutdown Panel

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER 05000
07	23	2017	2017	- 005	- 00	09	20	2017	FACILITY NAME	DOCKET NUMBER 05000

9. OPERATING MODE	12			
1	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
100 %	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
		☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT	TELEPHONE NUMER *(Include Area Code)*
Richard A. Loeffler, Senior Licensing Engineer	763-295-1247

13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
B	LA	TS	S345	Y					

14. SUPPLEMENTAL REPORT EXPECTED		15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
☐ YES *(If yes, complete 15. EXPECTED SUBMISSION DATE)*	☒ NO				

ABSTRACT *(Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)*

On July 23, 2017, the Emergency Diesel Generator (EDG) lube oil cooler immersion heater temperature switch failed (in the on position) which together with residual heat from 12 EDG operation actuated the EDG engine temperature switch resulting in a valid start signal to the 12 EDG Emergency Service Water (ESW) System pump and transfer of pump control to the Alternate Shutdown System panel. In accordance with 10 CFR 50.73(a)(2)(iv)(A) an automatic actuation of the ESW System is reportable. The apparent cause of the 12 EDG hot engine condition was a failed temperature switch for the immersion heater in concert with operating the 12 EDG. From the time of the problem identification through the 12 EDG surveillance test, lube oil temperature was monitored by Operations and never exceeded the normal operating band when the EDG was in the shutdown ready-to-start configuration and during the EDG monthly surveillance run. Therefore, the 12 EDG remained operable. The immediate corrective action was replacement of the failed temperature switch. A long-term corrective action is to evaluate the immersion heater and associated control circuit to determine if the temperature switch design and preventative maintenance frequency are appropriate. Corrective action will be taken based on the results of this evaluation.

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NARRATIVE

EVENT DESCRIPTION

On July 19, 2017, at 1149 hours, it was identified that lube oil temperature for the 12 Emergency Diesel Generator (EDG) [EIIS: DG] had made a step change stabilizing at approximately 150°F. The step change indicated that the immersion heater was either not operating properly or was cycling abnormally. On July 20, 2017, at 1634 hours, after troubleshooting, it was identified that the temperature switch controlling the 12 EDG immersion heater had failed, causing it to be continuously on. It was determined that since this condition did not result in temperatures outside the normal operating band, no compensatory measures were required but additional monitoring by Operations was implemented.

On July 23, 2017, with the unit in Mode 1, at 100 percent rated thermal power, surveillance procedure 0187-02, “12 Emergency Diesel Generator / 12 ESW [Emergency Service Water System] Quarterly Pump and Valve Test,” was performed. After shutting down the 12 EDG in accordance with the surveillance procedure, at 1332 hours, Annunciator 8-C-21 (NO. 12 DIESEL ENG TROUBLE) was logged in the Control Room. An in-plant operator reported that Annunciator C94-A-13 (HOT ENGINE) was in alarm and could not be reset. This condition resulted in a valid start signal to the 12 EDG Emergency Service Water System [BI] (EDG-ESW) pump (which was already running) and an automatic transfer of the pump control to the Alternate Shutdown System (ASDS) panel.

EDG cooling water flows from the heat exchangers to a EDG lube oil system [LA] lube oil cooler. At the inlet to the lube oil cooler is a manifold containing an engine temperature switch and an immersion heater [EHTR] temperature switch (TS-7195) [TS]. If EDG engine water temperature reaches 190°F the engine temperature switch actuates the hot engine relay, which starts and transfers control of the 12 EDG-ESW pump to the ASDS panel. During EDG engine shutdown, water heated by a 15 KW immersion heater (controlled by TS-7195) circulates through the lube oil cooler by natural circulation to maintain oil temperature between 135°F and 165°F; 85°F is the required minimum temperature to ensure EDG fast start capability.

Investigation identified that the switch contacts for the immersion heater temperature switch (Manufacturer: Square D, Model No. GZW2-5102, Serial No. 40036996) had failed in the closed position, causing the immersion heater to be continuously on when the 12 EDG was shutdown in a standby state. While this condition did not raise temperature out of specification prior to EDG engine operation, it did add enough local heat to bring in the alarm after the engine was shutdown following a run due to the residual heat.

This condition activated the engine temperature switch, in turn actuating the hot engine relay, which resulted in an auto start signal to the already running 12 EDG-ESW pump, and an automatic transfer of pump control to the ASDS panel. The immersion heater temperature switch was replaced on July 25, 2017.

EVENT ANALYSIS

This condition is being reported in accordance with 10 CFR 50.73(a)(2)(iv)(A), "Any event or condition that resulted in manual or automatic actuation of any of the systems listed in paragraph 10 CFR 50.73(a)(2)(iv)(B)". Specifically, criteria 10 CFR 50.73(a)(2)(iv)(B)(9) for ESW "systems that do not normally run and that serve as ultimate heat sinks" applies. This event is not classified as a safety system functional failure.

CAUSE

The apparent cause of the 12 EDG hot engine condition was a failed temperature switch for the immersion heater in concert with operating the 12 EDG. Following completion of the surveillance test, the 12 EDG was shutdown and returned to the automatic standby mode. When the EDG is shutdown the immersion heater automatically cycles back on and controls lube oil temperature to between 135°F and 165°F. With the immersion heater failed in the on position along with the residual heat from engine operation a localized hot spot occurred that was sufficient to actuate the transfer of control of the 12 EDG-ESW pump to the ASDS panel.

SAFETY SIGNIFICANCE

The immersion heater (and control) is required to maintain EDG engine operability while in the standby state by maintaining EDG lube oil temperature greater than 85°F to ensure fast start capability. With the immersion heater continuously on, the 12 EDG lube oil temperature stabilized at 154°F, well above the minimum and within the required band. Prior to the monthly surveillance test, engine temperatures recorded on operator rounds were within the normal acceptance band.

During the 12 EDG engine run, the coolant temperature was maintained at approximately 175°F, which is within the normal operating range, and below the hot engine alarm setpoint. From the time of problem identification through the 12 EDG surveillance test, lube oil temperature was monitored by operations and never exceeded the normal operating band. This includes when the EDG was in the shutdown ready-to-start configuration and during the EDG monthly surveillance run. Therefore, the 12 EDG remained operable.

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There was no safety significance to transmittal of the valid start signal to the 12 EDG-ESW pump. The start of the 12 EDG-ESW pump and transfer of control to the ASDS panel does not prevent the cooling of the 12 EDG or impact the ability of the 12 EDG to perform its safety function. This condition did not impact EDG, EDG-ESW System, or ASDS System operability since the lube oil temperatures never exceeded the normal EDG operating band, and the system logic operated as designed and engine cooling was always available.

CORRECTIVE ACTIONS

The immediate corrective action was to replace the failed immersion heater temperature switch, TS-7195.

The long-term corrective actions are:

1. Evaluate the immersion heater and associated control circuit to determine if the temperature switch design and preventative maintenance frequency are appropriate. Based on the results, take action to address identified deficiencies.

2. Revise plant procedures to include discussion of component(s) not operating correctly or in configurations that are abnormal, so that the impact of the condition is properly understood and controlled.

PREVIOUS SIMILAR EVENTS

On April 29, 2014, it was identified that a similar issue occurred following shutdown from performance of the surveillance. The engine temperature switch and the immersion heater temperature switch setpoint bands were found to be out of adjustment resulting in the 12 EDG engine residual heat actuating the hot engine temperature switch. This resulted in an automatic start signal to the 12 EDG-ESW pump and transfer of pump control to the ASDS panel. Note, the immersion heater temperature switch did not fail during this occurrence. The condition was corrected by installation of new temperature switches with the proper operating parameters.